Proposal for a collaborative project in response to the special call on

Digital Health in Developing Countries

by Else Kröner Fresenius Foundation

Project title: Telehealth Services for Myanmar (Telekyanmar Project)

B Persons and institutions

1. Persons

Project leader submitting the application

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Dr. Ulrich Kuch is the Head of the Department of Tropical Medicine and Public Health in the Institute for Occupational Medicine, Social Medicine and Environmental Medicine of the Health Sciences Centre, Faculty of Medicine of Goethe University in Frankfurt am Main, Germany.

A keen herpetologist and snake breeding expert since childhood, Dr. Kuch has more than 30 years of research experience at the interface of scientific-technical and development cooperation and the coordination of large inter- and transdisciplinary collaborative R&D projects. He has spent more than 8 years living and working abroad, mostly in snakebite endemic low- and middle income countries, conducting basic, applied and operational research on this and other neglected tropical diseases and their associated biotic and abiotic factors in close collaboration with local, national and international academics, governments and non-government organizations.

After studying genetics, toxicology, and zoology and being awarded several prestigious scholarships from organizations like the German Academic Scholarship Foundation, German Academic Exchange Service, Australian Research Council and Australian National University, he obtained the degree of a 'Diplom-Biologe' (MSc by research thesis equivalent) from Goethe University in 2000 and a PhD *summa cum laude* from the Life Sciences Department of Goethe University in 2007 for interdisciplinary research on the molecular biology, toxicology, taxonomy, evolution and medical importance of the South and Southeast Asian venomous snake genus *Bungarus* (kraits).

While a Graduate Research Assistant in the Departments of Forensic Genetics and Forensic Toxicology of Goethe University's Centre for Legal Medicine he worked in several research

consortia like "Isolation and characterization of marine bioactive compounds" (2000-2002, funded by the Federal Ministry of Education and Research of Germany) and "Novel ligands of ion channels from animal venoms" (2002-2004, funded by the German Science Foundation DFG) and invented a new, forensic trace DNA based method for diagnosing the species of snake involved in snakebites. Also during his PhD research, he participated as a Co-Applicant and Co-Investigator in a in Wellcome Trust funded research project on krait bites in Bangladesh (2006-2009; grant 079027/Z/06/Z).

From 2007 to 2008 Dr. Kuch worked as a Postdoctoral Scientific Assistant for Project Development reporting to the Director General of Germany's largest biodiversity research institution, the Senckenberg Gesellschaft für Naturforschung. He then worked as Project Group Leader for "Diversity and dynamics of pathogens, vectors and reservoir hosts under climate change conditions" (2008-2011) and Project Group Leader for "Medical Biodiversity Research" (2011-2014) in the Biodiversity and Climate Research Centre (BiK-F) in Frankfurt am Main, a major new research institute funded under the LOEWE programme for scientific and economic excellence of the State of Hesse. In this excellence centre he also led a Junior Research Group and served as Head of the Emerging and Neglected Tropical Diseases Unit.

After returning to the Faculty of Medicine of Goethe University in 2014 as Head of the Department of Tropical Medicine and Public Health in the Institute of Occupational Medicine, Social Medicine and Environmental Medicine, Dr. Kuch also worked for MSF Switzerland (Doctors without Borders) as an Epidemiological Activity Manager for Snakebite in the Abyei Special Administrative Area of South Sudan (2016) and as a Snakebite Research and Policy Advisor in South Sudan (2017).

Dr. Kuch's research interests focusing on snakes, snake toxins and snakebite envenoming extend from biodiversity, basic and applied biochemistry over product development for diagnostics and therapeutics well into clinical research and public health. He has published more than 100 research papers mostly on snakes, snakebite envenoming, and poisoning, but also maintains an active research programme on vector-borne and zoonotic diseases, health risks in wildlife trade and other aspects of pandemic prevention, climate change and health, disaster preparedness, and humanitarian medicine.

He was the Consortium Coordinator and Principal Investigator of the BMBF funded FlutNetz project improving access to care for medical emergencies in flood disaster situations in Bangladesh (2020-2024), Lead Applicant and Principal Investigator of a project aiming to derive repellents against the most dangerous snakes of Myanmar from a natural predator of vipers (2019-2022; funded by Volkswagen Foundation), and has been a Co-Applicant and Co-Investigator in several other research projects on snakebite envenoming like the SNAKE-BYTE project ('Tackling the second deadliest Neglected Tropical Disease: predicting and reducing the impact of snakebite on human and animal health through interdisciplinary analyses of hotspots and access to care; 2018-2022; funded by Swiss National Fund for Science SNF), a project on the 'Use of antivenoms for snakebite envenoming under routine program conditions: an observational study from the MSF Hospital in Agok, South Sudan' (2017-2020; funded by MSF Switzerland), the EU FP7 project BITES ('Biodiversity Information Technology and Exchange for Snakebite management'; 2013-2017; a Marie Curie International Research Staff Exchange Scheme with India), two clinical projects on rapid diagnostic tests for snake bite envenoming (2013-2015; UBS Optimus Foundation Global Health Research Innovations), a randomized, double-blind clinical trial of two dose regimens of polyvalent antivenom for the treatment of snake bites with neurotoxic envenoming in Nepal (2011-2013; Swiss National Fund for Science SNF), and a study on the epidemiology and clinical management of snake bite envenoming in Laos (2013-2014; funded by Else Kröner-Fresenius Foundation).

Dr. Kuch was the Lead Applicant and Speaker of the Graduate School Coordinated PhD Programme "Biodiversity and global change effects on emerging and neglected tropical

diseases" at Goethe University Frankfurt (2013-2018; DAAD Structured PhD Programme for University Graduate Schools) and has been a founding member of the Société Africaine de Venimologie (SAV/African Society of Toxinology), a founding member of the German Alliance for Global Health Research and the One Health Platform of Germany, the founder and Co-Chair of the Snakebite Envenoming Working Group of the Global Health Hub Germany, and a Life Member of the Toxicology Society of Bangladesh and the Bangladesh Society of Infectious and Tropical Diseases.

He regularly serves as an Associate Editor for PLoS Neglected Tropical Diseases and was a member of the editorial boards of other international journals. He often reviews papers for scientific journals and grant applications for funding agencies like the Wellcome Trust, NSF, Alexander von Humboldt Foundation and the Federal Ministry of Education and Research, and evaluates promotion and tenure documents for research institutes and universities worldwide. He was also appointed by WHO as the final reviewer of the WHO Guidelines on Production, Control and Regulation of Snake Antivenom Immunoglobulins (2010).

Dr. Kuch has conducted research in collaboration with partners in Myanmar since his first visit to this country in 2003, has spent several years in total living and working there, and has been the focal point of university partnerships between Goethe University Frankfurt and universities in Myanmar. Before the military coup of February 2021 in Myanmar, he carried out research on medically relevant biodiversity, social and biological research on the prevention of snakebite, and clinical research on the diagnosis and therapy of snakebite envenoming in Myanmar. A human rights activist since early youth, Dr. Kuch is also a member of German Solidarity with Myanmar Democracy e.V. and a frequent contributor to science-policy dialogue on related matters.

Following the military coup and the brutal persecution and killing of peaceful pro-democracy demonstrators in Myanmar, he and his team stopped or relocated their ongoing research projects and suspended partnerships with institutions that had fallen under the control of the Myanmar military. Instead, the team of Goethe University Frankfurt have since successfully engaged in the humanitarian assistance and medical development cooperation efforts for Myanmar by starting partnerships with hospitals in liberated areas of northern and south-eastern Myanmar that are administered by ethnic health organizations.

Four of these partnerships, with hospitals operated by the Karen Department of Health and Welfare (KDHW), have been supported by the GIZ funding programme Hospital Partnerships which is financed by the Federal Ministry for Economic Cooperation and Development of Germany and Else Kröner-Fresenius Foundation. In a new research project on pandemic preparedness, started in 2022 and funded by the Federal Ministry for Economic Cooperation and Development, Dr. Kuch and his team have also investigated health risks in the illegal wildlife trade in northern and south-eastern Myanmar in cooperation with ethnic health organizations and the Ministry of Health and the Ministry of Natural Resources and Environmental Conservation of the democratic National Unity Government of Myanmar. In a parallel development, Goethe University Frankfurt has started a new university partnership with Kachin State Comprehensive University (KSCU) and its School of Medicine and School of Nursing in 2024. KSCU is the first full university of the new federal education system of Myanmar that teaches in presence in Laiza (Kachin State).

In the successfully concluded (2020-2024), BMBF funded FlutNetz project which Dr. Kuch led as the Consortium Coordinator and Principal Investigator for Goethe University Frankfurt, one of the work packages of Goethe University was dedicated to establishing a novel telemedicine facility with coordinating, public awareness, health education, provider-to-patient and provider-to-provider capacities in Bangladesh. This facility has recently been handed over to the Ministry of Health and Family Welfare of Bangladesh that plans to expand it to become the National Poison Centre of Bangladesh.

Thus, the project proposal presented here is part of a broader cooperation in humanitarian assistance, higher education and biomedical research between Goethe University Frankfurt and democratic organizations and their institutions in Myanmar, and it builds on specific experiences with the establishment and operation of telemedicine services in fragile contexts gained by the team of Goethe University in Myanmar's neighbouring country of Bangladesh.

In this project, Dr. Kuch will fulfil the function of the Project Manager at Goethe University Frankfurt. He will have the overall responsibility for the project and will use 25% of his regular working time for reporting to EKFS, liaising with the participating ethnic health organizations, the Telekyanmar team, the Ministry of Health of the National Unity Government of Myanmar and other stakeholders, contributing to project coordination, management, and M&E on site in Myanmar, consultations on snakebite cases, data analysis, and the publication of project results. Salary support for this part of his duties is requested from EKFS.

Project leader for Myanmar

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Dr. Laxus Diego is an experienced medical doctor and technical advisor for health information systems, information security and data visualization. He holds an M.B., B.S. degree from the University of Medicine Magway in Myanmar (specialisations/majors: Medicine, Surgery, gynaecology and obstetrics, paediatrics) and completed a Tropical Medicine, Community Healthcare and Research intensive course at Khon Kaen University, Thailand, with a scholarship funded by TICA.

He was also awarded a prestigious Asia Pacific New Zealand Aid Scholarship for a Master Degree supported by New Zealand Ministry of Foreign Affairs and Trades, graduated with first class honours as a Master of Health Informatics from Auckland University of Technology (specialisations/majors: Clinical Information System and Electronic Health Records, Health Informatics and Health Sectors, Research Methods, Information Security, Health Analytics, Health Information Standards, Ubiquitous Computing and Usage Centred Design), and received the Te Ara Auaha – Facility of Design and Creative Technologies Dean's Award for Excellence in Postgraduate Study at Auckland University of Technology.

After working as a Medical Officer (emergency and primary responder, inpatient ward team leader and part of the surgical operation team) in Myat Taw Win Private Hospital of Taunggyi, Myanmar, Dr. Laxus Diego joined the Department of Public Health of the Ministry of Health and Sports of Myanmar as a Public Health Officer. There, he was first responsible for the registration, configuration, apps installation and distribution of over 12,000 mobile tablets to healthcare providers in 7 states and regions of Myanmar and monitoring them using a mobile device management system, for curriculum development, training and technical support for the states and regional public health departments in Myanmar, the development of job aid mobile

apps in collaboration with NGOs and INGOs, and for the administration of the Ministry's Website, the Department Facebook Page, and the Ministry's Mail Server.

Later, he was responsible for developing and revising the national eHealth architecture blueprint for Myanmar in cooperation with a consultant from the World Bank, and for developing and implementing a COVID-19 Database management system, which was used by over 1200 hospitals, quarantine centres and COVID-19 healthcare centres in Myanmar to record the data of every COVID-19 test and quarantined individual. During the third wave of COVID-19 in Myanmar, around 450 providers worked together and extended telehealth services to over 65,000 patients through COVID-19 specific telemedicine channels.

Dr. Laxus Diego is currently the Program Manager and IT Technical Lead of the Telekyanmar Telemedicine program and has provided over 180,000 consultations online within the past 3 years. At Telekyanmar, he has been responsible for the formulation, implementation, operation and evaluation of the telemedicine program including HR recruitment and training. With currently around 300 volunteers working together on this program, he and his team have successfully implemented the entire process of establishing the telemedicine program and providing healthcare services purely online.

Prof. Dr. David Groneberg is the tenured University Professor (W3) and Director of the Institute for Occupational, Social and Environmental Medicine at Goethe University Frankfurt. He previously served as a tenured University Professor (W2) and Director of the Institute for Occupational Medicine at Charité – University Medicine Berlin, a tenured University Professor (W2) in the Department of Pneumology at Hannover Medical School, a Research Group Leader at Charité University Medicine Berlin, and as a Junior Professor (W1) at the Leibniz Research Center Borstel and the University of Lübeck.

Prof. Groneberg is a Member of the Medical Scientific Board of the German Ministry of Defense, Referee for the Wissenschaftsrat of Germany, a Member of the Control Commission of the Federal Institute for Medical and Pharmaceutical Proficiency Assessment, and an Extraordinary Member of the Drug Commission of the German Medical Association. Among numerous academic distinctions, he was awarded the Prize of the Central Research Institute of Ambulatory Health Care in Germany (with Jan Bauer, 2016), the Fritz Lickint Medal of the German Society for Nicotine and Tobacco Research (2014), the Walther-Siegenthaler Prize (with Jan Bauer, 2014), the Research Prize of the German Society for Pediatric Infectiology (2002), and the Prize of the Hesse Chamber of Physicians (2002).

He and his team in the Institute for Occupational, Social and Environmental Medicine will contribute to this project by advising on cases in their respective medical specialities, on public health, access to healthcare, and health system reform, by providing working space and administrative support at Goethe University Frankfurt, and by participating in data analysis, publication of project results, and reporting.

For all persons for whom (partial) salaries are requested, exact tasks are explained in the Excel file detailing the project budget (part E of the application files).

2. Partner organisations

Goethe University Frankfurt (applicant organization)

Goethe University (GU) was founded in 1914 as a unique "citizens' university" in Frankfurt. Home to over 41,000 students, including about 8,000 international students from 150 countries, it is the third largest university in Germany. GU maintains partnerships and

collaborative projects with 700 universities all over the world. In keeping with its mission to support civic society, GU was one of the first higher education institutions in Germany to establish a dedicated academic pathway programme for refugees. The Faculty of Medicine of GU and University Hospital Frankfurt (UKF) with over 1300 physicians and scientists and about 2700 students in more than 50 specialist clinics, clinical, and research institutes ensure the integration of teaching, research, and patient care. We combine the special responsibility of training future generations of healthcare professionals with the opportunities created by integrating evidence-based scientific findings directly into patient care.

Telekyanmar Telemedicine Organization

Telekyanmar was established as a telemedicine program in 2021 by the Ministry of Health (MOH) of the National Unity Government of Myanmar (NUG) with the goal of providing essential medical services to people who faced challenges accessing healthcare during the current multidimensional crisis in Myanmar. Telekyanmar is now a registered NGO in Myanmar. Its mission is ensuring accessible healthcare for all, adapting to the evolving needs and circumstances in Myanmar in order to continue serving those who rely on telemedicine services for their well-being, and to build a sustainable model for future health service delivery across all of Myanmar. Telekyanmar, being an already established and operational telemedicine program, has achieved significant success in improving healthcare access for remote communities in Myanmar since June 2021. It has successfully provided more than 180,000 of remote consultations, facilitated timely diagnosis, and strengthened local healthcare capacity through training programs. Collaborating with ethnic health organisations in Myanmar, the program has overcome geographical, cultural, religious and linguistic barriers, ensuring that quality healthcare reaches underserved areas. With a proven track record of positive health outcomes and strong community engagement, Telekyanmar continues to expand its reach, demonstrating the potential of telemedicine to transform healthcare delivery across Myanmar's diverse regions.

3. Cooperation partners

Karen Department of Health and Welfare

The Karen Department of Health and Welfare (KDHW) is an ethnic health organization (EHO) dedicated to serving the health and welfare needs of the Karen population in Myanmar. Established in 1956, KDHW provides comprehensive healthcare services in south-eastern Myanmar, including primary and secondary care, through a network of 89 Village Tract Health Centers, 6 secondary care hospitals, 4 referral centers, 7 mobile clinics and 31 health posts. Its mission is to ensure that all individuals in the areas of Myanmar that are administered by the Karen National Union (KNU) areas have access to quality, affordable, and equitable health and welfare services. KDHW focuses on community-based healthcare, addressing common diseases, maternal and child health, emergency obstetric care, and health education.

Health Department, Kachin Political Interim Coordination Team

The Health Department of the Kachin Political Interim Coordination Team (KPICT) operates as part of a broader effort to represent and coordinate the political aspirations of the Kachin people in Myanmar. Formed in March 2021, KPICT aims to end authoritarianism and establish a genuine Federal Democratic Union in Myanmar. The Health Department of KPICT focuses on providing essential healthcare services to the population in Kachin State of northern Myanmar, particularly in areas affected by conflict and displacement. These initiatives include

distributing medical supplies, managing emergency health situations, and collaborating with both domestic and international organizations to improve health outcomes.

Health Department, Karenni State Interim Executive Council

The Health Department of the Karenni State Interim Executive Council (IEC) is a key component of the provisional government established in the Karenni State of Myanmar in June 2023. The Karenni State IEC was created to unify various Karenni resistance groups in their defence against the military junta of Myanmar and to provide stable governance in resistance-held areas. The Health Department of the Karenni State IEC is responsible for delivering essential healthcare services, including primary care, maternal and child health, and emergency medical responses. Its goal is to ensure that all residents of Karenni State have access to necessary health services, thereby improving overall health and well-being in the region.

Ministry of Health, National Unity Government of the Republic of Union of Myanmar

The interim National Unity Government of the Republic of the Union of Myanmar (NUG) is the only and legitimate Government of the Republic of the Union of Myanmar. After the illegal military coup of February 2021, the NUG was formed by the Committee Representing Pyidaungsu Hluttaw (CRPH) in discussion with the National Unity Consultative Council (NUCC) of Myanmar. The CRPH consists of members of the parliament of Myanmar who had been elected by the people of Myanmar in the last free, fair and democratic election held in this country in November 2020. The NUG implements the goal, objective and steps of the political road map for Myanmar as prescribed in Part I of the Federal Democracy Charter in discussion with the NUCC, and is accountable to the CRPH. The Ministry of Health (MOH) of the NUG is responsible for healthcare delivery and public health at the federal/union level in Myanmar and for the transformation of health policies and the health system towards federalism according to the Federal Democracy Charter. Its goal is to ensure that the people of Myanmar have access to the health services they need during the interim period of the ongoing anti-dictatorship movement by the people, which aims for the return of the power to the people, genuine freedom, federal democracy, equality, rule of law, and sustainable peace. The MOH follows its Strategic Healthcare Plan, the two cornerstones of which are Universal Health Coverage (UHC) and Federal Democratic Principles.